

ST JOHN'S INTL, NL, Canada

WMO: 718010

Lat: 47.620N

Lon: 52.750W

Elev: 141

StdP: 99.65

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.7	-12.0	-18.7	0.7	-12.8	-16.8	0.9	-11.3	19.0	-3.6	17.8	-3.0	7.1	290	0.826

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	25.0	19.3	23.5	18.5	22.1	17.8	20.8	23.2	19.8	22.2	18.7	21.1	6.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	14.9	22.2	18.8	13.8	21.3	17.8	13.0	20.2	60.6	23.4	57.3	22.3	53.8	21.1	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	13.9	12.5	-15.9	28.0	2.2	1.2	-17.4	28.9	-18.7	29.6	-19.9	30.3	-21.5	31.2
			-16.5	22.5	2.1	0.9	-18.0	23.1	-19.3	23.6	-20.5	24.1	-22.0	24.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-3.5	-4.2	-2.2	2.0	6.2	10.8	16.4	16.6	12.8	7.9	3.6	-1.0
	DBStd	8.13	4.22	4.39	3.85	3.44	3.74	4.13	3.55	3.04	3.45	3.57	3.86	4.13
	HDD10.0	2244	420	397	378	242	131	39	2	1	12	86	196	341
	HDD18.3	4721	678	630	637	491	377	228	79	69	169	323	442	598
	CDD10.0	604	0	0	0	1	13	63	201	205	96	22	3	1
	CDD18.3	38	0	0	0	0	0	2	18	16	3	0	0	0
	CDH23.3	143	0	0	0	0	1	7	76	52	8	0	0	0
	CDH26.7	8	0	0	0	0	0	0	4	3	1	0	0	0
(14) Wind	WSAvg	6.2	7.3	7.2	6.8	6.3	5.6	5.4	5.5	5.2	5.6	6.0	6.5	7.0
(15) Precipitation	PrecAvg	1547	155	137	138	118	103	98	86	117	124	161	152	157
	PrecMax	1929	238	245	223	297	183	186	176	337	256	329	284	266
	PrecMin	1180	68	62	58	36	22	24	23	44	34	82	55	64
	PrecStd	178	49	43	42	56	41	40	37	61	48	53	51	52
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	9.7	10.3	15.7	21.2	24.4	27.2	26.8	24.3	19.9	16.4	12.8
		MCWB	10.7	8.9	8.6	12.1	15.2	18.2	20.4	20.6	19.5	18.0	15.1	12.0
	2%	DB	7.6	6.4	7.4	12.3	18.1	22.3	25.3	24.8	21.7	17.3	14.1	9.9
		MCWB	7.2	5.9	5.9	9.5	13.3	16.7	19.3	19.5	18.2	15.6	13.2	9.4
	5%	DB	4.3	3.8	4.7	9.7	15.5	20.1	23.9	23.2	20.0	15.5	12.1	6.9
		MCWB	3.9	3.1	3.4	7.5	11.6	15.6	18.8	18.4	17.1	14.1	11.2	6.5
	10%	DB	2.0	1.6	2.7	7.3	12.9	18.0	22.4	21.8	18.3	13.6	9.8	4.5
		MCWB	1.4	0.9	1.5	5.4	9.7	14.3	18.0	17.8	15.9	12.0	9.0	4.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	9.1	8.9	13.4	16.8	20.0	22.3	22.1	21.0	18.3	15.4	12.3
		MCDB	11.3	9.7	10.2	14.9	19.4	22.9	24.9	25.0	22.6	19.3	16.0	12.8
	2%	WB	7.2	5.9	6.3	10.3	14.3	18.4	21.0	20.7	19.1	16.1	13.3	9.4
		MCDB	7.6	6.4	7.4	11.5	16.8	20.8	23.7	23.1	20.9	16.9	13.9	9.8
	5%	WB	3.9	3.3	3.6	7.7	12.1	16.6	19.9	19.6	17.6	14.2	11.2	6.5
		MCDB	4.2	3.7	4.3	9.1	14.6	18.8	22.6	22.0	19.4	15.2	12.0	6.8
	10%	WB	1.5	1.0	1.7	5.5	10.0	14.8	18.8	18.5	16.3	12.1	9.1	4.1
		MCDB	1.9	1.4	2.4	7.1	12.7	17.4	21.3	20.9	18.0	13.3	9.8	4.4
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.3	6.1	6.8	8.3	8.7	8.8	7.7	7.6	6.1	6.0	5.5
		MCDBR	10.5	9.4	9.0	10.9	12.9	12.3	10.7	9.9	9.4	8.6	9.1	9.2
		MCWBR	10.7	9.5	8.0	8.5	8.7	7.6	6.3	5.8	6.2	7.3	8.8	9.5
	5% WB	MCDBR	10.4	9.4	8.8	10.5	11.6	10.9	9.4	8.8	8.7	8.5	9.0	9.3
		MCWBR	10.7	9.6	8.5	9.1	8.8	7.9	6.4	6.2	6.8	7.8	8.9	9.6
(40) Clear-Sky Solar Irradiance	taub		0.315	0.315	0.334	0.365	0.368	0.401	0.429	0.396	0.369	0.344	0.323	0.312
	taud		2.359	2.433	2.413	2.370	2.424	2.356	2.317	2.421	2.469	2.493	2.463	2.373
	Ebn at Noon		755	851	888	888	895	864	832	847	840	806	748	710
	Edh at Noon		72	82	98	113	111	120	123	106	93	77	65	63
(44) All-Sky Solar Radiation	RadAvg		0.94	1.72	2.88	3.94	4.81	5.09	5.12	4.33	3.44	2.08	1.18	0.78
	RadStd		0.07	0.13	0.23	0.39	0.39	0.40	0.38	0.28	0.22	0.19	0.16	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page